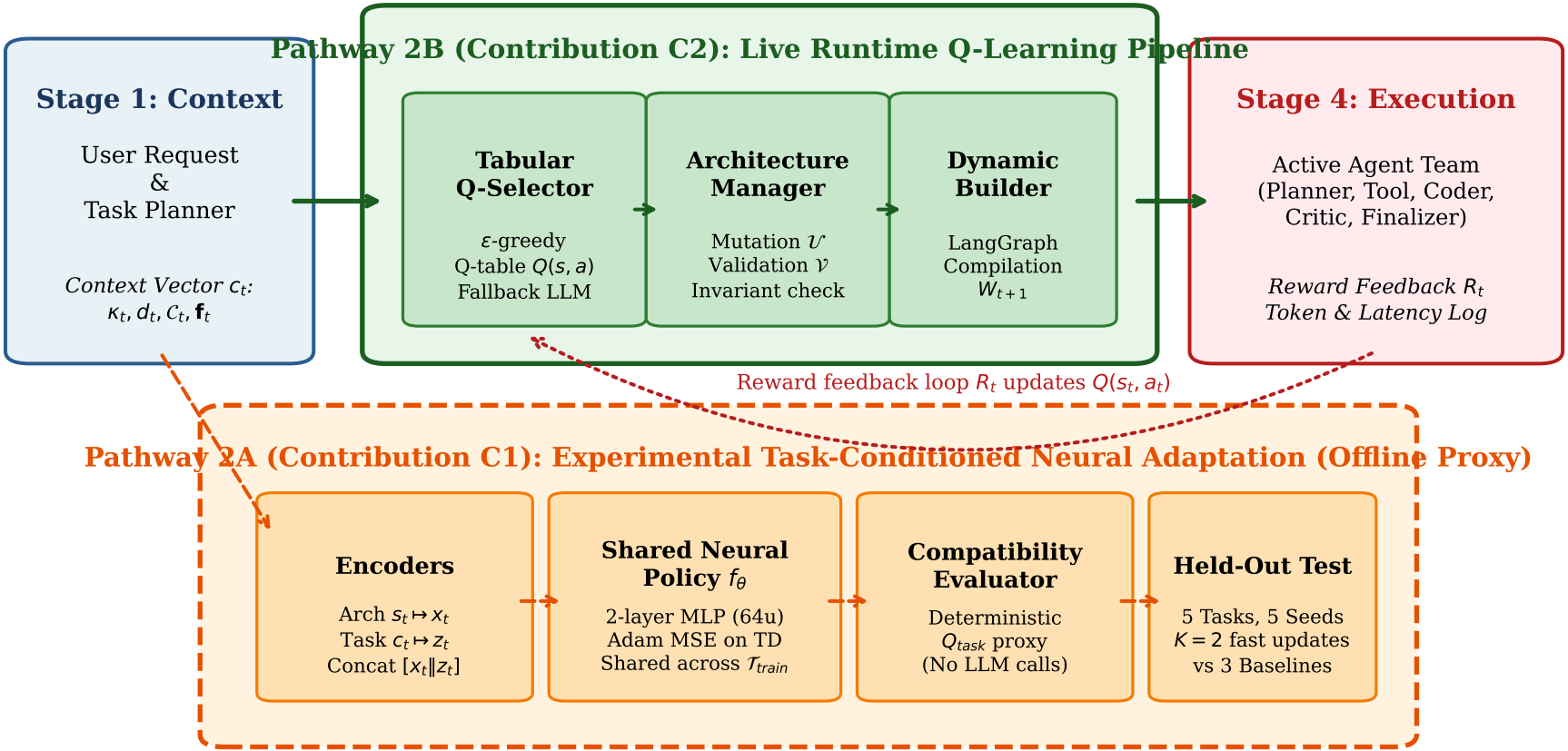

Pathway 2B (Contribution C2): Live Runtime Q-Learning Pipeline
Stage 1: Context
User Request
&
Task Planner
Context Vector c_t:
$\kappa_t, d_t, C_t, \mathbf{f}_t$
Tabular
Q-Selector
ε-greedy
Q-table $Q(s, a)$
Fallback LLM
Architecture
Manager
Mutation $\mathcal{U}$
Validation $\mathcal{V}$
Invariant check
Dynamic
Builder
LangGraph
Compilation
W_{t+1}
Stage 4: Execution
Active Agent Team
(Planner, Tool, Coder,
Critic, Finalizer)
Reward Feedback R_t
Token & Latency Log
Reward feedback loop R_t updates $Q(s_t, a_t)$
Pathway 2A (Contribution C1): Experimental Task-Conditioned Neural Adaptation (Offline Proxy)
Encoders
Arch $s_t \mapsto x_t$
Task $c_t \mapsto z_t$
Concat $[x_t \| z_t]$
Shared Neural
Policy f_θ
2-layer MLP (64u)
Adam MSE on TD
Shared across $\mathcal{T}_{train}$
Compatibility
Evaluator
Deterministic
Q_{task} proxy
(No LLM calls)
Held-Out Test
5 Tasks, 5 Seeds
$K = 2$ fast updates
vs 3 Baselines